Getting Ready for Similarity

Similar rectangles and similar triangles

Scale factor is always written as 2nd figure ÷ 1st figure.

Part A Similar Rectangles

EASY

1. Long: $\frac{6}{3} = 2$, Short: $\frac{4}{2} = 2 \Rightarrow$ **Similar**, scale factor 2

2. Long: $\frac{12}{4} = 3$, Short: $\frac{9}{3} = 3 \Rightarrow$ **Similar**, scale factor 3

3. Long: $\frac{7}{5} = \frac{7}{5}$, Short: $\frac{4}{2} = 2 \Rightarrow$ **Not similar**

4. Long: $\frac{12}{6} = 2$, Short: $\frac{8}{4} = 2 \Rightarrow$ **Similar**, scale factor 2

5. Long: $\frac{8}{5} = \frac{8}{5}$, Short: $\frac{6}{3} = 2 \Rightarrow$ **Not similar**

6. **M1:** Scale factor $\frac{6}{2} = 3$; $x = 4 \times 3 = 12$

M2: $\frac{6}{2} = \frac{x}{4} \Rightarrow 2x = 6 \times 4 = 24 \Rightarrow x = 24 \div 2 = 12$

7. **M1:** Scale factor $\frac{14}{7} = 2$; $x = 3 \times 2 = 6$

M2: $\frac{14}{7} = \frac{x}{3} \Rightarrow 7x = 14 \times 3 = 42 \Rightarrow x = 42 \div 7 = 6$

8. **M1:** Scale factor $\frac{10}{5} = 2$; $x = 2 \times 2 = 4$

M2: $\frac{10}{5} = \frac{x}{2} \Rightarrow 5x = 10 \times 2 = 20 \Rightarrow x = 20 \div 5 = 4$

9. **M1:** Scale factor $\frac{12}{4} = 3$; $x = 9 \times 3 = 27$

M2: $\frac{12}{4} = \frac{x}{9} \Rightarrow 4x = 12 \times 9 = 108 \Rightarrow x = 108 \div 4 = 27$

10. **M1:** Scale factor $\frac{15}{5} = 3$; $x = 6 \times 3 = 18$

M2: $\frac{15}{5} = \frac{x}{6} \Rightarrow 5x = 15 \times 6 = 90 \Rightarrow x = 90 \div 5 = 18$

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MEDIUM

1. Long: $\frac{9}{6} = \frac{3}{2}$, Short: $\frac{6}{4} = \frac{3}{2} \Rightarrow$ **Similar**, scale factor $\frac{3}{2}$

2. Long: $\frac{14}{12} = \frac{7}{6}$, Short: $\frac{10}{8} = \frac{5}{4} \Rightarrow$ **Not similar**

3. Long: $\frac{12}{8} = \frac{3}{2}$, Short: $\frac{9}{6} = \frac{3}{2} \Rightarrow$ **Similar**, scale factor $\frac{3}{2}$ (2nd rectangle is turned sideways)

4. Long: $\frac{15}{10} = \frac{3}{2}$, Short: $\frac{8}{4} = 2 \Rightarrow$ **Not similar**

5. Long: $\frac{6}{9} = \frac{2}{3}$, Short: $\frac{4}{6} = \frac{2}{3} \Rightarrow$ **Similar**, scale factor $\frac{2}{3}$

6. **M1:** Scale factor $\frac{10}{4} = \frac{5}{2}$; $x = 6 \times \frac{5}{2} = 15$

M2: $\frac{10}{4} = \frac{x}{6} \Rightarrow 4x = 10 \times 6 = 60 \Rightarrow x = 60 \div 4 = 15$

7. **M1:** Scale factor $\frac{6}{9} = \frac{2}{3}$; $x = 6 \times \frac{2}{3} = 4$

M2: $\frac{6}{9} = \frac{x}{6} \Rightarrow 9x = 6 \times 6 = 36 \Rightarrow x = 36 \div 9 = 4$

8. **M1:** Scale factor $\frac{18}{12} = \frac{3}{2}$; $x = 8 \times \frac{3}{2} = 12$

M2: $\frac{18}{12} = \frac{x}{8} \Rightarrow 12x = 18 \times 8 = 144 \Rightarrow x = 144 \div 12 = 12$

9. **M1:** Scale factor $\frac{6}{15} = \frac{2}{5}$; $x = 10 \times \frac{2}{5} = 4$

M2: $\frac{6}{15} = \frac{x}{10} \Rightarrow 15x = 6 \times 10 = 60 \Rightarrow x = 60 \div 15 = 4$

10. **M1:** Scale factor $\frac{25}{10} = \frac{5}{2}$; $x = 4 \times \frac{5}{2} = 10$

M2: $\frac{25}{10} = \frac{x}{4} \Rightarrow 10x = 25 \times 4 = 100 \Rightarrow x = 100 \div 10 = 10$

Part B Similar Triangles

EASY

1. $\frac{4}{2} = 2$, $\frac{6}{3} = 2$, $\frac{8}{4} = 2 \Rightarrow \triangle ABC \sim \triangle DEF$, scale factor 2

2. $\frac{9}{3} = 3$, $\frac{15}{5} = 3$, $\frac{21}{7} = 3 \Rightarrow \triangle ABC \sim \triangle PQR$, scale factor 3

3. $\frac{6}{4} = \frac{3}{2}$, $\frac{7}{5} = \frac{7}{5}$, $\frac{8}{6} = \frac{4}{3} \Rightarrow$ **Not similar**

4. $\frac{10}{5} = 2$, $\frac{12}{6} = 2$, $\frac{16}{8} = 2 \Rightarrow \triangle ABC \sim \triangle JKL$, scale factor 2

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5. $\frac{6}{3} = 2, \quad \frac{8}{4} = 2, \quad \frac{10}{6} = \frac{5}{3} \Rightarrow$ **Not similar**

6. **M1:** $AB \leftrightarrow DE$: $\frac{12}{4} = 3$; $x = EF$ matches $BC = 5$: $x = 5 \times 3 = 15$

M2: $\frac{DE}{AB} = \frac{EF}{BC}$: $\frac{12}{4} = \frac{x}{5} \Rightarrow 4x = 12 \times 5 = 60 \Rightarrow x = 60 \div 4 = 15$

7. **M1:** $AB \leftrightarrow PQ$: $\frac{8}{4} = 2$; $x = RP$ matches $CA = 3$: $x = 3 \times 2 = 6$

M2: $\frac{PQ}{AB} = \frac{RP}{CA}$: $\frac{8}{4} = \frac{x}{3} \Rightarrow 4x = 8 \times 3 = 24 \Rightarrow x = 24 \div 4 = 6$

8. **M1:** $BC \leftrightarrow YZ$: $\frac{14}{7} = 2$; $x = XY$ matches $AB = 6$: $x = 6 \times 2 = 12$

M2: $\frac{YZ}{BC} = \frac{XY}{AB}$: $\frac{14}{7} = \frac{x}{6} \Rightarrow 7x = 14 \times 6 = 84 \Rightarrow x = 84 \div 7 = 12$

9. **M1:** $AB \leftrightarrow JK$: $\frac{21}{7} = 3$; $x = KL$ matches $BC = 9$: $x = 9 \times 3 = 27$

M2: $\frac{JK}{AB} = \frac{KL}{BC}$: $\frac{21}{7} = \frac{x}{9} \Rightarrow 7x = 21 \times 9 = 189 \Rightarrow x = 189 \div 7 = 27$

10. **M1:** $AB \leftrightarrow ST$: $\frac{16}{8} = 2$; $x = US$ matches $CA = 6$: $x = 6 \times 2 = 12$

M2: $\frac{ST}{AB} = \frac{US}{CA}$: $\frac{16}{8} = \frac{x}{6} \Rightarrow 8x = 16 \times 6 = 96 \Rightarrow x = 96 \div 8 = 12$

MEDIUM

1. $\frac{6}{4} = \frac{3}{2}, \quad \frac{9}{6} = \frac{3}{2}, \quad \frac{12}{8} = \frac{3}{2} \Rightarrow \triangle ABC \sim \triangle QRP$, scale factor $\frac{3}{2}$

2. $\frac{3}{6} = \frac{1}{2}, \quad \frac{4}{8} = \frac{1}{2}, \quad \frac{5}{10} = \frac{1}{2} \Rightarrow \triangle ABC \sim \triangle ZXY$, scale factor $\frac{1}{2}$

3. $\frac{8}{6} = \frac{4}{3}, \quad \frac{12}{9} = \frac{4}{3}, \quad \frac{15}{12} = \frac{5}{4} \Rightarrow$ **Not similar**

4. $\frac{12}{8} = \frac{3}{2}, \quad \frac{15}{10} = \frac{3}{2}, \quad \frac{18}{12} = \frac{3}{2} \Rightarrow \triangle ABC \sim \triangle LJK$, scale factor $\frac{3}{2}$

5. $\frac{10}{5} = 2, \quad \frac{12}{6} = 2, \quad \frac{16}{9} = \frac{16}{9} \Rightarrow$ **Not similar**

6. **M1:** $AB \leftrightarrow DE$: $\frac{12}{8} = \frac{3}{2}$; $x = FD$ matches $CA = 10$: $x = 10 \times \frac{3}{2} = 15$

M2: $\frac{DE}{AB} = \frac{FD}{CA}$: $\frac{12}{8} = \frac{x}{10} \Rightarrow 8x = 12 \times 10 = 120 \Rightarrow x = 120 \div 8 = 15$

7. **M1:** $AB \leftrightarrow PQ$: $\frac{6}{9} = \frac{2}{3}$; $x = RP$ matches $CA = 12$: $x = 12 \times \frac{2}{3} = 8$

M2: $\frac{PQ}{AB} = \frac{RP}{CA}$: $\frac{6}{9} = \frac{x}{12} \Rightarrow 9x = 6 \times 12 = 72 \Rightarrow x = 72 \div 9 = 8$

8. **M1:** $AB \leftrightarrow YZ$: $\frac{10}{4} = \frac{5}{2}$; $x = XY$ matches $CA = 8$: $x = 8 \times \frac{5}{2} = 20$

M2: $\frac{YZ}{AB} = \frac{XY}{CA}$: $\frac{10}{4} = \frac{x}{8} \Rightarrow 4x = 10 \times 8 = 80 \Rightarrow x = 80 \div 4 = 20$

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9. **M1:** $AB \leftrightarrow KL$: $\frac{6}{8} = \frac{3}{4}$; $x = LJ$ matches $BC = 12$: $x = 12 \times \frac{3}{4} = 9$

M2: $\frac{KL}{AB} = \frac{LJ}{BC}$: $\frac{6}{8} = \frac{x}{12} \Rightarrow 8x = 6 \times 12 = 72 \Rightarrow x = 72 \div 8 = 9$

10. **M1:** $BC \leftrightarrow ST$: $\frac{15}{10} = \frac{3}{2}$; $x = AB$ matches $RS = 18$: $x = 18 \div \frac{3}{2} = 12$

M2: $\frac{ST}{BC} = \frac{RS}{AB}$: $\frac{15}{10} = \frac{18}{x} \Rightarrow 15x = 10 \times 18 = 180 \Rightarrow x = 180 \div 15 = 12$

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